Domain-Specific Tokenization for Biomedical NLP

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Project Access

Repository:
notebooks + tokenizers + poster code

Study topic:
domain-specific tokenization

Motivation

- Biomedical text contains long compounds, abbreviations, and rare terms that are hard for generic tokenizers.
- Over-fragmentation increases sequence length, weakens entity boundaries, and may harm minority classes.
- Goal: train compact domain BPE tokenizers and test whether improved segmentation translates to downstream gains.

Dataset Signals (PubMedQA)

Corpus Profile

Labeled: 1,000 **Unlabeled:** 61,249
Question length: mean 12.9 words
Context length: mean 200.2 words
Class balance: yes 55.2%, no 33.8%, maybe 11.0%

- Contexts dominate tokenization cost and drive model efficiency.
- Minority class (*maybe*) motivates macro-F1 over accuracy-only evaluation.

Experimental Setup

Baselines

BERT-base | BioBERT | GPT-2

Custom

BioBPE-30k | BioBPE-50k

Protocol: 300 samples for intrinsic analysis; PubMedQA 70/15/15 for classification; JNLPBA 4k/1k for NER metrics.

Hypothesis

Domain-trained BPE will reduce fragmentation in biomedical text and improve task-level robustness, especially for entity-heavy inputs.

Tokenizer Adaptation Pipeline

1) Collect PubMedQA corpora
↓
2) Train biomedical BPE vocabularies (30k/50k)
↓
3) Evaluate intrinsic metrics (fertility, compactness, UNK)
↓
4) Validate on classification and NER-oriented metrics
↓
5) Rank tokenizers by efficiency + quality trade-off

Qualitative term behavior: baseline tokenizers split terms like *gastroesophageal* into 5+ pieces, while custom BPE often keeps the whole term in one token.

Tokenizer Adaptation Methods

FVT (Fast Vocabulary Transfer)

$$E_{\text{new}}(t_i) = \frac{1}{|T_d(t_i)|} \sum_{t_j \in T_d(t_i)} E_{\text{old}}(t_j)$$

Initializes each new token embedding as the average of embeddings of its source subpieces from the old vocabulary.

FOCUS

$$E_{\text{new}}(t_i) = \frac{1}{|V_t|} \sum_{t_j \in V_t} w_t \, E_{\text{old}}(t_j)$$

Uses weighted averaging with similarity-aware coefficients to better preserve semantic neighborhoods.

ZeTT (Zero-shot Tokenizer Transfer)

$$\mathcal{L}_\phi = \mathcal{L}_{\text{LM}} + \alpha \cdot \mathcal{L}_{\text{aux}}$$

Learns tokenizer-adaptation parameters by combining language modeling loss with tokenizer-specific auxiliary constraints.

Evaluation Signals

- **Segmentation compactness:** avg tokens per sample, fertility, token length.
- **Downstream utility:** test accuracy and macro-F1 on yes/no/maybe.
- **NER compatibility:** subtoken-per-entity ratio, single-piece entity rate, fragmentation rate.

Composite NER tokenization score

$$\text{Score} = 0.50(1 - \text{Frag}) + 0.30(\text{SinglePiece}) + 0.20(1 - \text{UNK})$$

Experimental Results

Stage 4: Intrinsic tokenization metrics (best to worst by fertility)

Tokenizer	Avg Tok/Sample	Fertility	Avg Tok Len
BioBPE-50k	273.26	**1.301**	5.12
BioBPE-30k	275.11	1.310	5.08
GPT-2	313.38	1.492	5.01
BERT-base	323.89	1.542	4.29
BioBERT	339.49	1.616	4.05

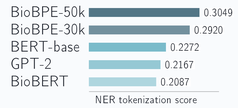
Relative gain: best custom model reduces fertility by ∼12.8% vs best baseline (GPT-2).

Stage 5: Downstream classification (test set)

Tokenizer	Accuracy	Macro-F1
BioBPE-50k	**0.5667**	**0.3718**
BioBPE-30k	**0.5667**	0.3710
GPT-2	**0.5667**	0.3643
BERT-base	0.5600	0.3582
BioBERT	0.5533	0.3521

Observation: custom BPE improves macro-F1 while keeping accuracy stable.

Stage 6: NER-oriented ranking



Key Takeaways

- Best tokenizer overall: **BioBPE-50k**.
- Intrinsic compactness and extrinsic quality improve consistently.
- Gains are strongest in entity-sensitive biomedical text.

Conclusions

- Domain-adapted BPE produces consistently more compact segmentation on biomedical text.
- Custom tokenizers deliver the best macro-F1 and strongest NER-oriented scores.
- 50k vocabulary is the most stable option across intrinsic and extrinsic evaluations.
- Practical recommendation: use custom biomedical BPE as a default tokenizer for PubMed-style pipelines.

Limitations: fixed lightweight classifier and single-domain corpus; future work includes transformer fine-tuning with embedding transfer (FVT/FOCUS/ZeTT).

Reproducibility

- Notebooks: 6 staged experiments from EDA to NER tokenization evaluation.
- Deterministic setup: fixed seeds and consistent data protocol.
- All metrics reported on the same benchmark splits for fair comparison.